

Lab Assignment 8

CS 331: Machine Learning

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March 17, 2025. To be done individually.

This lab will be about perceptron algorithm for classification.

1. Generate a linearly separable dataset in $\mathbb{R}^2$ with 100 points. There are several ways to do this. One way is to generate points from two well-separated Gaussian distributions.
2. Implement perceptron algorithm learnt in class. Provide the following plots:
 - (a) Single plot containing initial decision surface and final decision surface along with training data
 - (b) Objective function $J(\mathbf{w})$ that you are optimizing vs iterations.
3. Implement batch perceptron algorithm and provide the same plots as above. The difference from (2) is that, instead of making an update with respect to a single misclassified point, overall gradient of the objective function is used with respect to all the misclassified points.